



KING KHALID UNIVERSITY
Department of Computer Science
College of Computer Science

Parallel and Distributed Computing– (482-CCS-3)

Dr. Mohammad Nadeem Ahmed

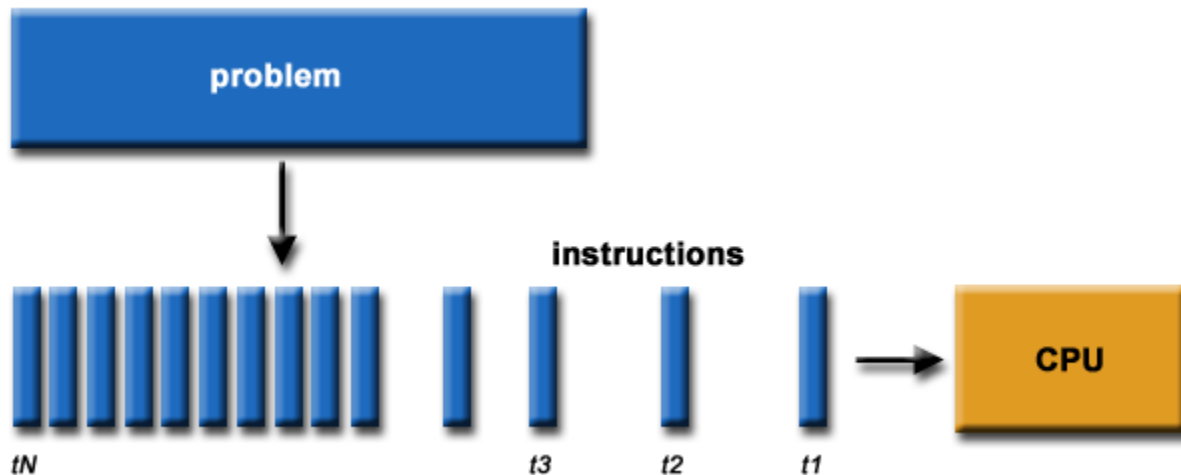
Chapter 6

Introduction to Parallel Computing: What and why is parallel computing, Serial (Sequential) Computing, Big Data ,5 V's of data, Why Big Data, Big Data in Real World.

Introduction to Parallel Computing

Traditionally, software has been written for *serial* computation:

- To be run on a single computer having a single Central Processing Unit (CPU);
- A problem is broken into a discrete series of instructions.
- Instructions are executed one after another.
- Only one instruction may execute at any moment in time.



A real-life example of this would be people standing in a queue waiting for a movie ticket and there is only a cashier. The cashier is giving tickets one by one to the persons. The complexity of this situation increases when there are 2 queues and only one cashier

In the simplest sense, parallel computing is the simultaneous use of multiple compute resources to solve a computational problem.

- To be run using multiple CPUs
- A problem is broken into discrete parts that can be solved concurrently
- Each part is further broken down to a series of instructions

Parallel Computing resources: -

The compute resources can include:

- A single computer with multiple processors;
- A single computer with (multiple) processor(s) and some specialized computer resources (GPU, FPGA ...)
- An arbitrary number of computers connected by a network;
- A combination of both.

Some Example:

- Planetary and galactic orbits
- Weather and ocean patterns
- Tectonic plate drift
- Rush hour traffic in Paris
- Automobile assembly line
- Daily operations within a business
- Building a shopping mall
- Ordering a hamburger at the drive through.
- weather and climate
- chemical and nuclear reactions
- biological, human genome
- geological, seismic activity
- mechanical devices - from prosthetics to spacecraft
- electronic circuits
- manufacturing processes

Serial (Sequential)

Serial (Sequential) Computing is a computational process in which tasks are executed one after another in a linear sequence. This means that each instruction or operation is processed individually, and only one task is being executed at any given time on a single processor. In serial computing, tasks are completed step-by-step, moving to the next step only after the current one is finished.

Example:-

Downloading Files

Single-threaded Web Browsers

Audio or Video Processing

Data Encryption

Text-Based Games

Limitations of Serial Computing

- Limits to serial computing - both physical and practical reasons pose significant constraints to simply building ever faster serial computers.
- Transmission speeds - the speed of a serial computer is directly dependent upon how fast data can move through hardware. Absolute limits are the speed of light (30 cm/nanosecond) and the transmission limit of copper wire (9 cm/nanosecond). Increasing speeds necessitate increasing proximity of processing elements.
- Limits to miniaturization - processor technology is allowing an increasing number of transistors to be placed on a chip. However, even with molecular or atomic-level components, a limit will be reached on how small components can be.
- Economic limitations - it is increasingly expensive to make a single processor faster. Using a larger number of moderately fast commodity processors to achieve the same (or better) performance is less expensive.

WHY PARALLEL COMPUTING

- Save time - wall clock time
- Solve larger problems
- Provide concurrency (do multiple things at the same time)
- Taking advantage of non-local resources - using available compute resources on a wide area network, or even the Internet when local compute resources are scarce.
- Cost savings - using multiple "cheap" computing resources instead of paying for time on a supercomputer.
- Overcoming memory constraints - single computers have very finite memory resources. For large problems, using the memories of multiple computers may overcome this obstacle.
- Ensures the effective utilization of the resources. The hardware is guaranteed to be used effectively whereas in serial computation only some part of the hardware was used and the rest rendered idle.

Future of Parallel Computing

The computational graph has undergone a great transition from serial computing to parallel computing. Tech giant such as Intel has already taken a step towards parallel computing by employing multicore processors. Parallel computation will revolutionize the way computers work in the future, for the better good. With all the world connecting to each other even more than before, Parallel Computing does a better role in helping us stay that way. With faster networks, distributed systems, and multi-processor computers, it becomes even more necessary.

- during the past 10 years, the trends indicated by ever faster networks, distributed systems, and multi-processor computer architectures (even at the desktop level) clearly show that *parallelism is the future of computing*.
- It will be multi-forms, mixing general purpose solutions (your PC...) and very speciliazed solutions as IBM Cells, ClearSpeed, GPGPU from NVidia ...

What is Big Data

Big data is a collection of large datasets that cannot be processed using traditional computing techniques. It is not a single technique or a tool, rather it has become a complete subject, which involves various tools, techniques and frameworks.

What Comes Under Big Data?

Big data involves the data produced by different devices and applications. Given below are some of the fields that come under the umbrella of Big Data.

- **Black Box Data** – It is a component of helicopter, airplanes, and jets, etc. It captures voices of the flight crew, recordings of microphones and earphones, and the performance information of the aircraft.
- **Social Media Data** – Social media such as Facebook and Twitter hold information and the views posted by millions of people across the globe.
- **Stock Exchange Data** – The stock exchange data holds information about the ‘buy’ and ‘sell’ decisions made on a share of different companies made by the customers.
- **Power Grid Data** – The power grid data holds information consumed by a particular node with respect to a base station.
- **Transport Data** – Transport data includes model, capacity, distance and availability of a vehicle.
- **Search Engine Data** – Search engines retrieve lots of data from different databases

5V's of Big Data

- **Velocity**: The data is increasing at a very fast rate. It is estimated that the volume of data will double in every 2 years.
- **Variety**: Now a days data are not stored in rows and column. Data is structured as well as unstructured. Log file, CCTV footage is unstructured

data. Data which can be saved in tables are structured data like the transaction data of the bank.

- Volume: The amount of data which we deal with is of very large size of Peta bytes.
- Veracity: inconsistencies and uncertainty in data.
- Value: meaning full data after data processing.

1. Volume:

- ❖ The name 'Big Data' itself is related to a size which is enormous. Volume is a huge amount of data.
- ❖ To determine the value of data, size of data plays a very crucial role. If the volume of data is very large, then it is actually considered as a 'Big Data'. This means whether a particular data can actually be considered as a Big Data or not, is dependent upon the volume of data.
- ❖ Hence while dealing with Big Data it is necessary to consider a characteristic 'Volume'.
- ❖ Example: In the year 2016, the estimated global mobile traffic was 6.2 Exabytes (6.2 billion GB) per month. Also, by the year 2020 we will have almost 40000 Exabytes of data.

2. Velocity:

- ❖ Velocity refers to the high speed of accumulation of data.
- ❖ In Big Data velocity data flows in from sources like machines, networks, social media, mobile phones etc.
- ❖ There is a massive and continuous flow of data. This determines the potential of data that how fast the data is generated and processed to meet the demands.
- ❖ Sampling data can help in dealing with the issue like 'velocity'.
- ❖ Example: There are more than 3.5 billion searches per day are made on Google. Also, Facebook users are increasing by 22%(Approx.) year by year.

3. Variety:

- ❖ It refers to nature of data that is structured, semi-structured and unstructured data.
- ❖ It also refers to heterogeneous sources.
- ❖ Variety is basically the arrival of data from new sources that are both inside and outside of an enterprise. It can be structured, semi-structured and unstructured.
- ❖ Structured data: This data is basically an organized data. It generally refers to data that has defined the length and format of data.
- ❖ Semi- Structured data: This data is basically a semi-organised data. It is generally a form of data that do not conform to the formal structure of data. Log files are the examples of this type of data.
- ❖ Unstructured data: This data basically refers to unorganized data. It generally refers to data that doesn't fit neatly into the traditional row and column structure of the relational database. Texts, pictures, videos etc. are the examples of unstructured data which can't be stored in the form of rows and columns.

4. Veracity:

- ❖ It refers to inconsistencies and uncertainty in data, that is data which is available can sometimes get messy and quality and accuracy are difficult to control.
- ❖ Big Data is also variable because of the multitude of data dimensions resulting from multiple disparate data types and sources.
- ❖ Example: Data in bulk could create confusion whereas less amount of data could convey half or Incomplete Information.

5. Value:

- ❖ After having the 4 V's into account there comes one more V which stands for Value! The bulk of Data having no Value is of no good to the company, unless you turn it into something useful.

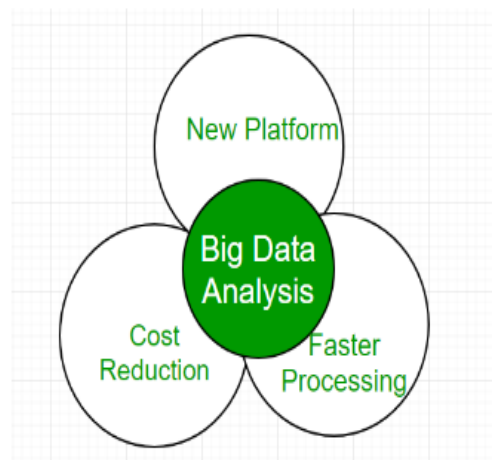
- ❖ Data in itself is of no use or importance but it needs to be converted into something valuable to extract Information. Hence, you can state that Value! is the most important V of all the 5V's

The data in it will be of three types.

- Structured data – Relational data.
- Semi Structured data – XML data,email,CSV,XML,JSON
- Unstructured data – Media Logs,audio,images

Why Big Data & Big Data in Real World.

Big data services enable a more comprehensive understanding of trends and patterns, by integrating diverse data sets to form a complete picture.



Big Data is used for many things but some of the main usage purposes are as follows:

- Big Data enables you to gather information about customers and their experience, then eventually helps you to align it properly.
- Helps in maintaining the predictive failures beforehand by analyzing the problems and provides with their potential solutions.
- Big Data is also useful for companies to anticipate customer demand, roll out new plans, test markets, etc.

- Big Data is very useful in assessing predictive failures by analyzing various indicators such as unstructured data, error messages, log entries, engine temperature, etc.
- Big Data is also very efficient in maintaining operational functions along with anticipating future demands of the customers, current market demands thus providing proper results.
- Hospitals are analyzing medical data and records to predict those patients that are likely to seek re-admission within a few months of discharge. The hospital can avoid costly stays for patients in the hospital.
- Financial Services organizations are using big data for data mining about customer interactions to slice and dice their users into finely tuned segments, this will help in creating increasingly relevant and sophisticated offers.



Advertisers and marketers—to tailor offers and promotions, and to make customer recommendations

Educational institutions—to minimize drop-outs, offer tailored learning, and to improve teaching methods

Healthcare providers—to create new treatments, develop wearable devices, and to improve clinical research

Transport and logistics—to streamline supply chain operations, improve airline safety, and even to save fuel and reduce carbon emissions

Banking and finance—to help prevent fraud, as well as to offer customers tailored products based on their activity

Agriculture—to help farmers perform as efficiently as possible and to monitor their crops